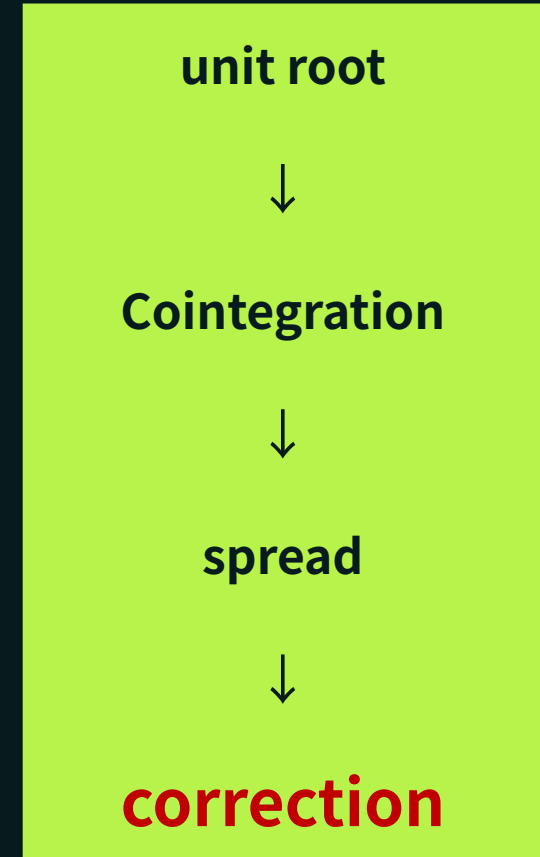


Prices that wander apart, one long-run equilibrium

From the cointegration test to the error-correction model



Four questions, one procedure

01 — Foundations

stationarity and cointegration

spurious regression · the I(d) ladder

02 — Engle–Granger

two assets, two steps

levels OLS · residual ADF · pairs trading

03 — Johansen

when there are several

VECM rank · Trace · three and five assets

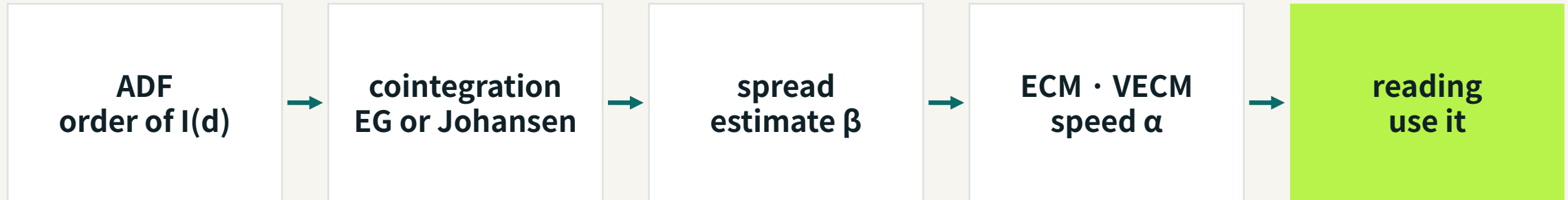
04 — **ECM**

how the gap closes

adjustment speed · weak exogeneity · uses

The test is not the goal; it is the gate to a spread and an adjustment speed

The whole procedure is five steps



DECISION RULE

Reject the null

COINTEGRATED

Steps 1 and 3 are unit-root tests; Step 2 is the regression that links them.

01

Foundations — stationarity and cointegration

Why a regression in levels is dangerous, and what replaces it

A stationary series comes back; a nonstationary one leaves

STATIONARY · $I(0)$

It comes back

- shocks fade away
- fixed mean and variance
- t and F tests valid

white noise, AR(1) with $|\phi| < 1$



NONSTATIONARY · $I(1)$

The level wanders

- shocks are permanent
- no mean to return to
- OLS inference breaks

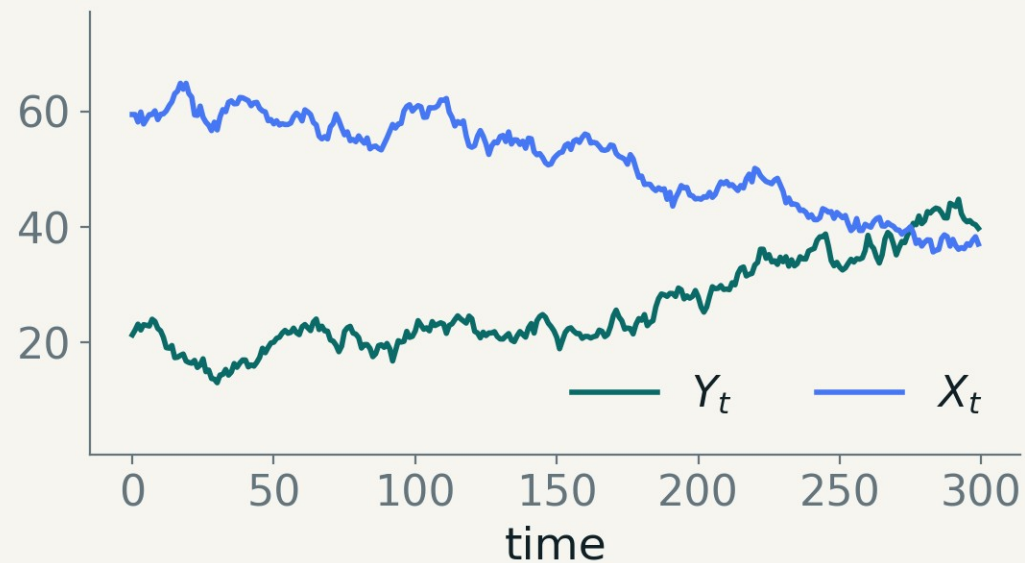
random walk, most asset prices

Differencing restores stationarity but throws the levels away.

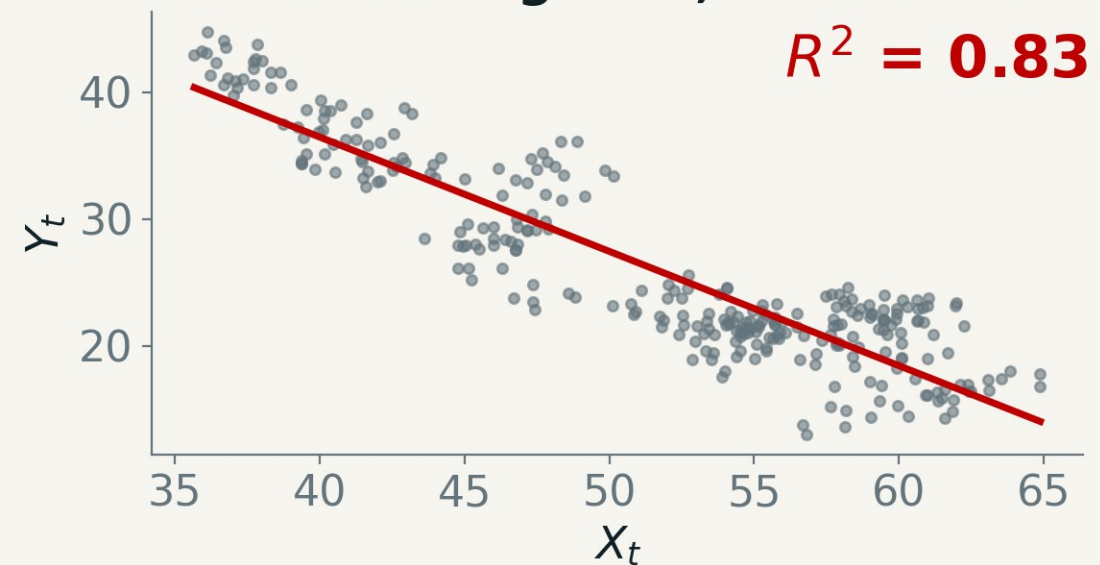
Cointegration keeps the levels and still gives valid inference.

Regressing two independent random walks invents a relationship

Two INDEPENDENT random walks



OLS in levels: tight fit, no relationship



While the residual wanders, a high R^2 and a large t-value prove nothing.

※ Simulated: two independent random walks, $T = 300$ — the residual is still $I(1)$

The order of integration counts the differences you need

$$\Delta^d Y_t \sim I(0) \iff Y_t \sim I(d)$$

$I(0)$

Already stationary

levels already revert

$I(1)$

One difference

the random-walk family

$I(2)$

two differences

breaks the recipe

The question only makes sense when both variables sit on the same rung, $I(1)$.

Cointegration is the claim that one linear combination is stationary

Definition

$$Y_t \sim I(1), \quad X_t \sim I(1) \quad \text{but} \quad Y_t - \beta X_t \sim I(0)$$

WHAT IT MEANS

A long-run tether

deviations are temporary

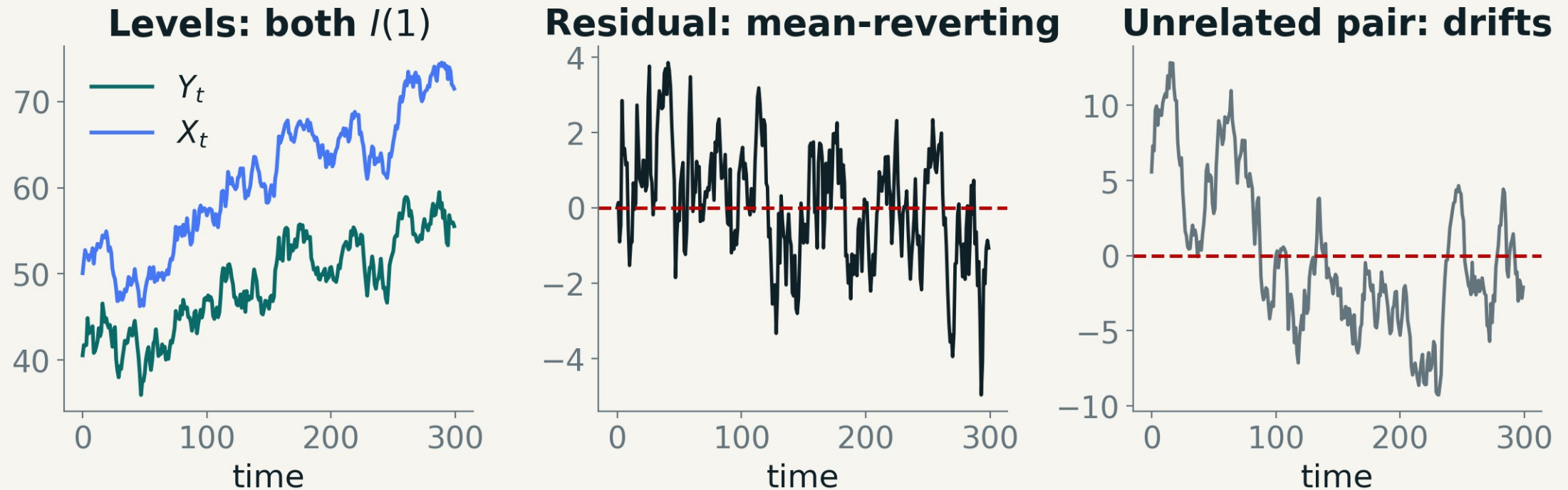
FOR TRADING

β is the hedge ratio

it is the position ratio

Individually $I(1)$, jointly $I(0)$ — that is the entire claim.

Cointegration shows up as a spread that returns to its mean



※ Simulated: Y and X are both $I(1)$; only the left pair leaves a stationary residual

02

Engle-Granger — two assets, two steps

Estimate the equilibrium in levels, judge it by the residual

Step 0 — check that each series is I(1)

ADF regression on the level

$$\Delta Y_t = \mu + \gamma Y_{t-1} + \sum_{i=1}^p \phi_i \Delta Y_{t-i} + \varepsilon_t$$

Hypotheses

$$H_0 : \gamma = 0 \text{ (unit root)} \quad H_1 : \gamma < 0 \text{ (stationary)}$$

Fail to reject in levels, reject in differences, and the series is I(1).

※ An I(2) series cannot be paired with an I(1) one. Add a trend only if the level clearly trends

Step 1 — OLS in levels gives the long-run coefficient

Cointegrating regression

$$Y_t = \alpha + \beta X_t + u_t$$

INTERCEPT

long-run level α

SLOPE

hedge ratio β

RESIDUAL

gap u

β converges fast thanks to super-consistency, but its t-value never decides cointegration.

Step 2 — the residual unit-root test decides

Residual ADF regression (no constant)

$$\Delta \hat{u}_t = \rho \hat{u}_{t-1} + \sum_{i=1}^p \gamma_i \Delta \hat{u}_{t-i} + \varepsilon_t$$

NULL

$$\rho = 0$$

I(1) residual

ALTERNATIVE

$$\rho < 0$$

I(0) residual, related

LAG p

AIC · BIC

whitening comes first

Rejection is evidence of cointegration; failing to reject proves nothing.

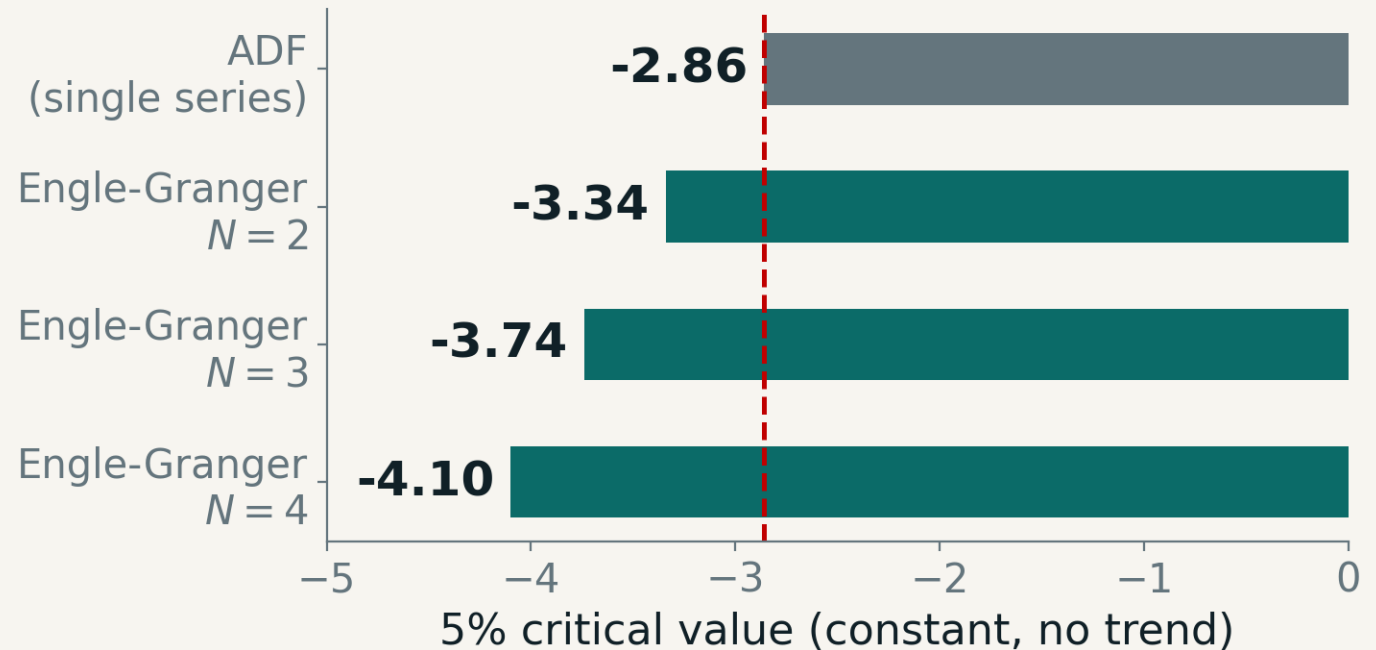
Ordinary ADF critical values over-reject here

CRITICAL VALUES

Shifted left

- the residual is estimated
- OLS flattens it
- MacKinnon tables

ADF tables over-reject



The more variables in the regression, the further left the rejection bar moves.

※ 5% values, constant and no trend. N counts the variables in the levels regression

A pairs trade is the residual, traded back to zero

Spread and its standardized score

$$\hat{u}_t = Y_t - \hat{\alpha} - \hat{\beta}X_t, \quad z_t = \frac{\hat{u}_t}{\hat{\sigma}_{\hat{u}}}$$

ENTRY

$z > +2$

short Y, buy β units of X

EXIT

$z \rightarrow 0$

equilibrium restored

STOP

$|z| > 3$

the relation may be gone

The traded object is the residual, and the hedge ratio comes straight out of Step 1.

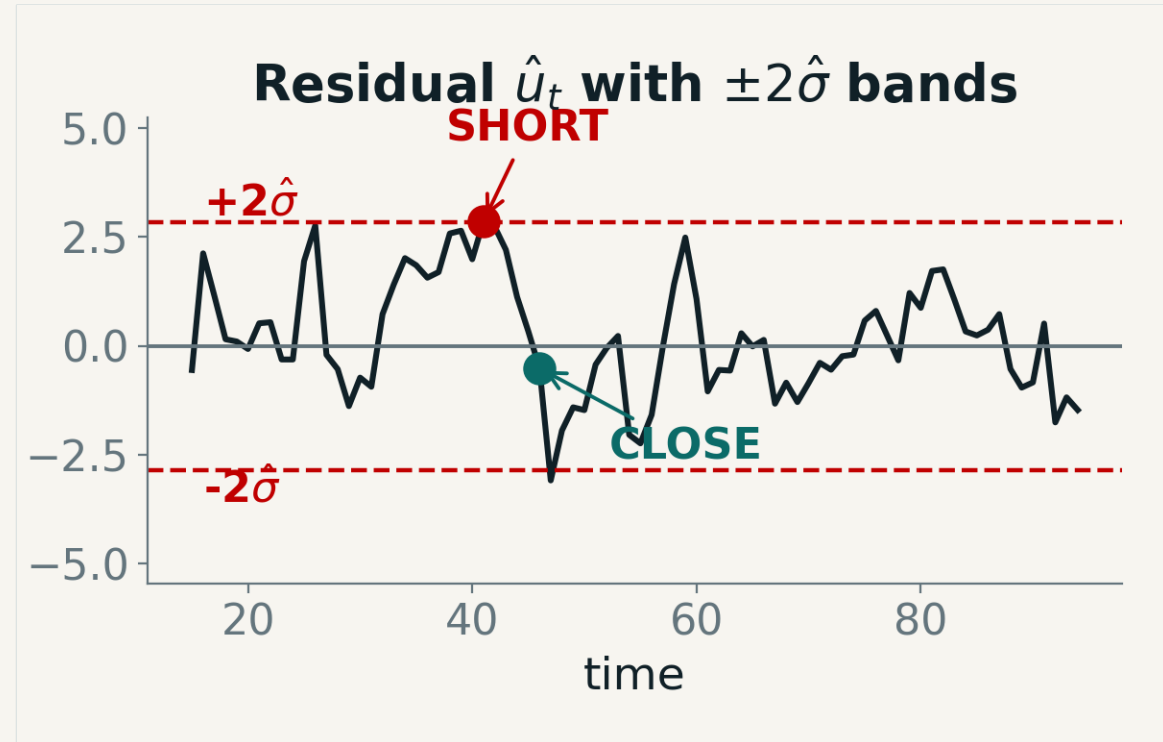
Example — short the spread at $+2\sigma$, close at the mean

ONE ROUND TRIP

Two-asset pair

- $\hat{\beta} = 0.67$
- $\hat{\sigma} = 1.42$
- enter at $z = +2.0$

held 5 periods



The return comes from mean reversion of the residual, not the direction of prices.

※ Simulated pair, $T = 300$. Entry and exit are the first band crossing after $t = 40$

The P&L of one round trip

hedge ratio $\hat{\beta}$

0.67

residual $\hat{\sigma}$

1.42

entry $\hat{u}$

+2.86

exit $\hat{u}$

-0.51

P&L on $Q = 100$ spread units

$$\Pi = \left(\hat{u}_{t_0} - \hat{u}_{t_1} \right) \times Q = \left(2.86 - (-0.51) \right) \times 100 \approx 337$$

Short 1 unit of Y against 0.67 of X: the gain is the shrinkage of the residual.

※ Illustrative: gross of transaction costs, financing and slippage

03

Johansen — when there is more than one relation

Test the rank of the long-run matrix instead of running a regression

Past three assets, two steps are not enough

Engle-Granger

One residual

- finds one relation only
- depends on the left-hand side
- the count stays unknown

fits two assets



Johansen

Rank test

- estimates how many
- order of variables irrelevant
- maximum likelihood

the default for three or more

Run EG on five assets and you get one residual with both relations mixed in.

The starting point is the VAR rewritten as a VECM

The long-run matrix, factored

$$\Delta Y_t = \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \mu + \varepsilon_t, \quad \Pi = \alpha \beta'$$

rank r	meaning	what to do
r = 0	no cointegration	VAR in differences
0 < r < k	r relations, k–r common trends	VECM
r = k	the series are stationary	VAR in levels

Only the long-run term is in levels, so $\beta'Y$ must be stationary for it to balance.

Estimation reduces to an eigenvalue problem

Generalized eigenvalue problem

$$\left| \lambda S_{11} - S_{10} S_{00}^{-1} S_{01} \right| = 0 \implies \hat{\lambda}_1 > \hat{\lambda}_2 > \dots > \hat{\lambda}_k$$

auxiliary regressions

$R_0 \cdot R_1$ residuals

lagged terms removed

moment matrices

four S matrices

not a correlation matrix

eigenvectors

estimate of β

the space only

A large eigenvalue is the signal that the combination is stationary.

Trace, maximum eigenvalue, and the sequential rule

Trace — are there at most r relations

$$LR_{tr}(r) = -T \sum_{i=r+1}^k \ln(1 - \hat{\lambda}_i)$$

Maximum eigenvalue — r against $r+1$

$$LR_{max}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$$

Start at $r = 0$ and take the first r you fail to reject.

The critical values are Johansen–Juselius, not chi-squared.

Three assets — one relation, two common trends

NULL	eigenvalue	Trace	5% critical	verdict
$r \leq 0$	0.109	62.27	29.80	reject
$r \leq 1$	0.009	4.99	15.49	fail to reject
$r \leq 2$	0.001	0.41	3.84	—

Spread normalized on C (true -0.5, -0.8)

$$\hat{S}_t = C_t - 0.468 A_t - 0.829 B_t$$

Half-life 2.4 periods. Enter at $t = 151$ with $z = +2.20$, close at $t = 155$ for $+4.57$.

Five assets — built from three common trends

asset	construction	character
A	$f_1 + \text{stationary}$	$I(1)$
B	$f_2 + \text{stationary}$	$I(1)$
C	$f_3 + \text{stationary}$	$I(1)$
D	$0.6f_1 + 0.4f_2 + \text{stationary}$	tied to A and B
E	$0.3f_2 + 0.9f_3 + \text{stationary}$	tied to B and C

The true relations are $D - 0.6A - 0.4B$ and $E - 0.3B - 0.9C$ — two of them.

Both statistics give the same answer — rank 2

NULL	eigenvalue	Trace	5%	max-eigen	5%	verdict
$r \leq 0$	0.116	145.1	69.8	74.0	33.9	reject
$r \leq 1$	0.076	71.2	47.9	47.6	27.6	reject
$r \leq 2$	0.020	23.6	29.8	12.0	21.1	fail to reject
$r \leq 3$	0.010	11.6	15.5	5.9	14.3	—
$r \leq 4$	0.009	5.6	3.8	5.6	3.8	—

The eigenvalues break sharply from 0.116 and 0.076 to 0.020. Three common trends.

Normalization separates the two relations

Relation 1 — normalized on D

$$\hat{S}_{1t} = D_t - 0.564A_t - 0.449B_t + 0.020C_t$$

Relation 2 — normalized on E

$$\hat{S}_{2t} = E_t - 0.023A_t - 0.301B_t - 0.917C_t$$

The two standardized spreads correlate at -0.04 .

Two nearly independent sources of return — something EG cannot deliver.

Four traps in Johansen

item	symptom	what to do
lag choice	rank overstated	AIC · BIC on the level VAR, put $p-1$ in
deterministics	r inflated on drifting data	set <code>det_order</code> from the data's trend
identifying β	only the space is identified	impose restrictions when $r \geq 2$
small samples	over-rejection	degrees-of-freedom correction $T \rightarrow T - kp$

Never read economic meaning straight off the top r eigenvectors.

04

ECM — how the gap closes

Reading the output of the test as a speed of adjustment

If there is cointegration, an error-correction form must exist

Every term balances at $I(0)$

$$\underbrace{\Delta Y_t}_{I(0)} = c + \lambda \underbrace{\hat{u}_{t-1}}_{I(0)} + \theta \underbrace{\Delta X_t}_{I(0)} + \varepsilon_t$$

VAR in differences

level information lost

long-horizon forecasts diverge

VAR in levels

No restriction

cannot state the long run

Rejecting in Step 2 is what licenses Step 3.

ECM holds short-run change and long-run repair in one equation

Single-equation ECM

$$\Delta Y_t = c + \lambda \hat{u}_{t-1} + \sum_{i=1}^p \phi_i \Delta Y_{t-i} + \sum_{j=0}^q \theta_j \Delta X_{t-j} + \varepsilon_t$$

DIFFERENCES
**short-run
dynamics**

immediate and lagged

ERROR-CORRECTION TERM

The gap

distance from balance

λ
**speed of
adjustment**

share closed per period

β was already estimated in Step 1 and is not re-estimated inside the ECM.

Read the speed by its sign, size and significance

$$t_{1/2} = \frac{\ln 2}{-\ln(1+\lambda)}$$

λ

reading

near 0, insignificant

no adjustment — revisit the verdict

$-1 < \lambda < 0$

monotone partial adjustment — the common case

$\lambda < -1$

overshooting — judge by the stability of the system

$\lambda > 0$, significant

divergence — misspecification or a sign problem

$\lambda = -0.25$ gives a half-life of 2.4 periods; $\lambda = -0.10$ gives 6.6.

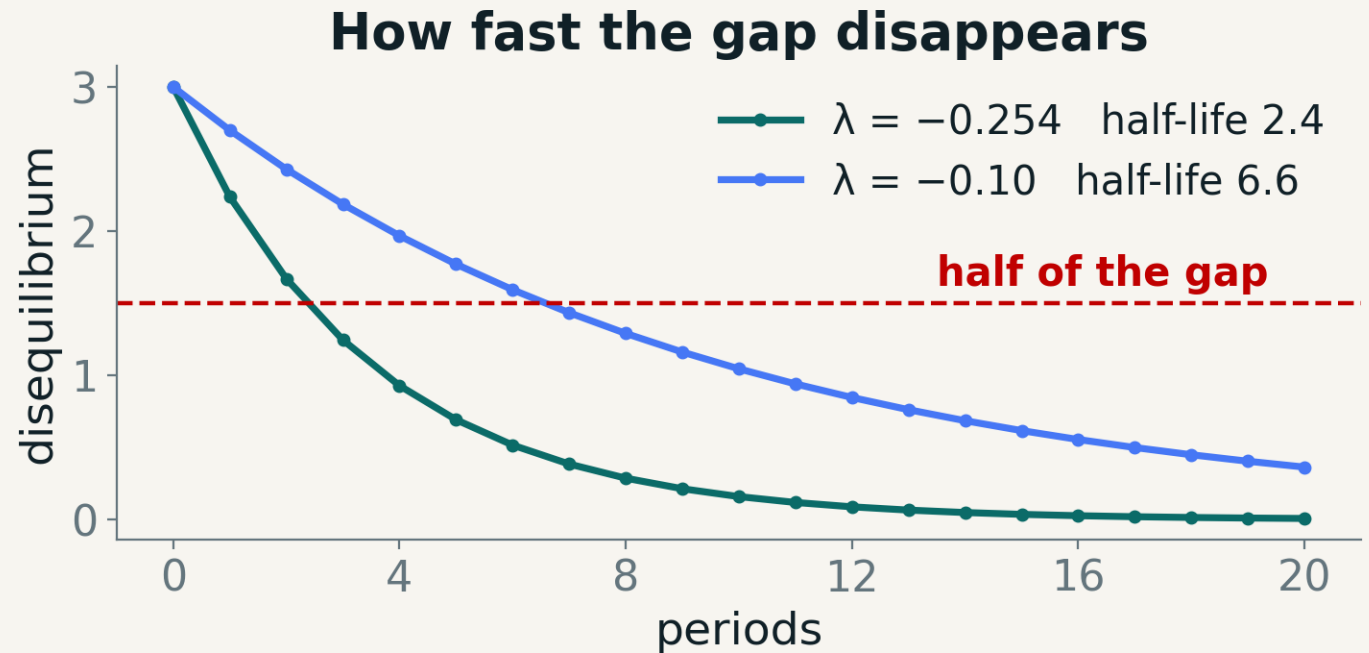
The same gap disappears very differently at different speeds

DECAY

Two half-lives

- initial gap 3.0
- $\lambda = -0.254$
- $\lambda = -0.10$

**two and a half days
versus a week and a
half**



The speed sets both the holding period of a trade and the convergence of a forecast.

Macro example — a one-off bonus to income

Long-run equilibrium — long-run MPC 0.9

$$C_t = 0.9 Y_t + u_t, \quad \hat{u}_{t-1} = C_{t-1} - 0.9 Y_{t-1}$$

ECM — short-run 0.4, speed -0.3

$$\Delta C_t = 0.4 \Delta Y_t - 0.3 \hat{u}_{t-1}$$

Start at equilibrium $Y = 100$, $C = 90$. A bonus of +20 at $t = 1$; income back to 100 from $t = 2$.

Income is a one-period step; consumption leaves a tail that runs for many.

※ A short-run coefficient below the long-run one is what consumption smoothing looks like

The first two periods, by hand

t = 1 — the bonus period

$$\Delta C_1 = 0.4(20) - 0.3(0) = +8 \Rightarrow C_1 = 98, \quad \hat{u}_1 = 98 - 108 = -10$$

t = 2 — income returns

$$\Delta C_2 = 0.4(-20) - 0.3(-10) = -5 \Rightarrow C_2 = 93, \quad \hat{u}_2 = 93 - 90 = +3$$

$$\hat{u}_t = 3(0.7)^{t-2}, \quad C_t = 90 + \hat{u}_t \quad (t \geq 2)$$

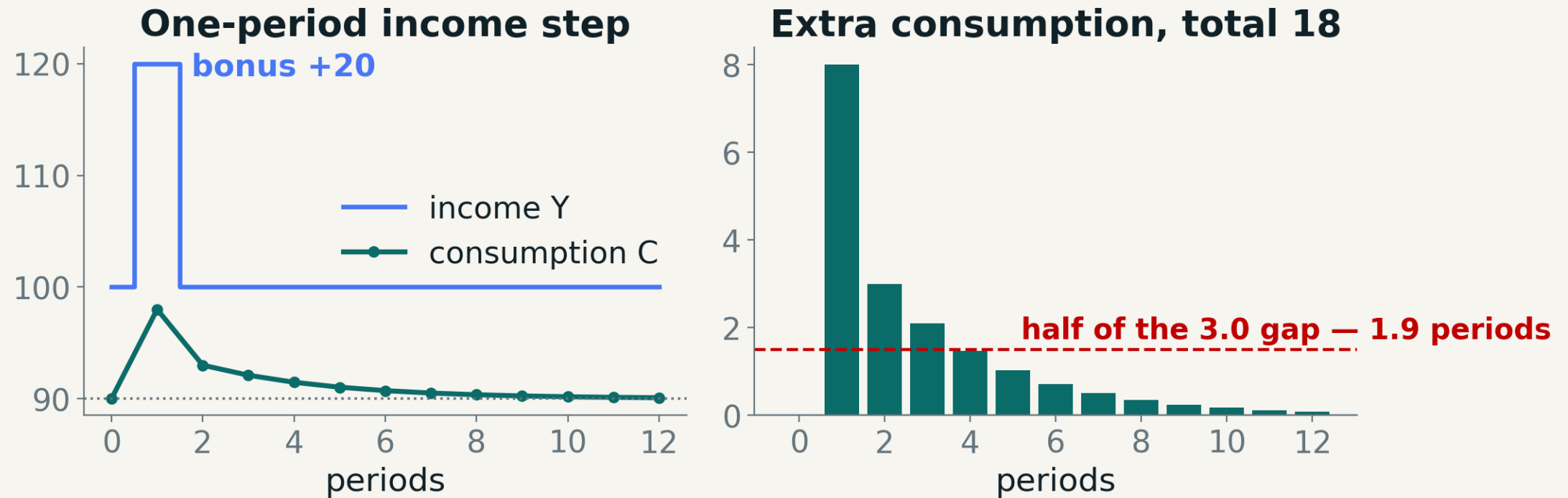
※ From t = 3 income is flat, so only the error-correction term is at work

The consumption path, period by period

t	income Y	ΔC	consumption C	gap $\hat{u}$
0	100	—	90.00	0.00
1	120	+8.00	98.00	−10.00
2	100	−5.00	93.00	+3.00
3	100	−0.90	92.10	+2.10
5	100	−0.44	91.03	+1.03
10	100	−0.07	90.17	+0.17

At t = 1 the gap is −10, below equilibrium; from t = 2 it is +3, coming down from above.

A step in income becomes an exponential tail in consumption



※ Right panel: extra consumption over the 90 baseline. The gap decays as $3 \times 0.7^{(t-2)}$

Half-life sets the speed, the long-run coefficient sets the total

Half-life — when half the gap is gone

$$t_{1/2} = \frac{\ln 2}{-\ln(1+\lambda)} = \frac{\ln 2}{-\ln(0.7)} = \frac{0.693}{0.357} = 1.9 \text{ periods}$$

Cumulative extra consumption

$$8 + \sum_{t \geq 2} 3(0.7)^{t-2} = 8 + \frac{3}{1-0.7} = 18 = 0.9 \times 20$$

The short run decides when it is spent; the long-run coefficient decides how much.

※ Under a strict permanent-income view the short-run coefficient is near 0; 0.4 signals liquidity constraints

Two-asset ECM — 25% of the gap closes in one period

Step 1 long-run fit (true 3, 1.5)

$$\hat{Y}_t = 1.47 + 1.536 X_t + \hat{u}_t, \quad \hat{\sigma}_{\hat{u}} = 1.52$$

Step 3 ECM (constant omitted)

$$\Delta Y_t = -0.254 \hat{u}_{t-1} + 1.473 \Delta X_t + 0.011 \Delta Y_{t-1}$$

speed λ

-0.254

t-value

-7.55

half-life

2.4 per

R^2

0.686

Cointegration test -7.56 ($p = 0.000$); short-run pass-through 1.473 ($t = 28.0$).

Who adjusts — the weak exogeneity test

dependent variable	coefficient on the gap	t-value	verdict
ΔY	−0.303	−4.93	adjusts
ΔX	−0.022	−0.66	does not adjust

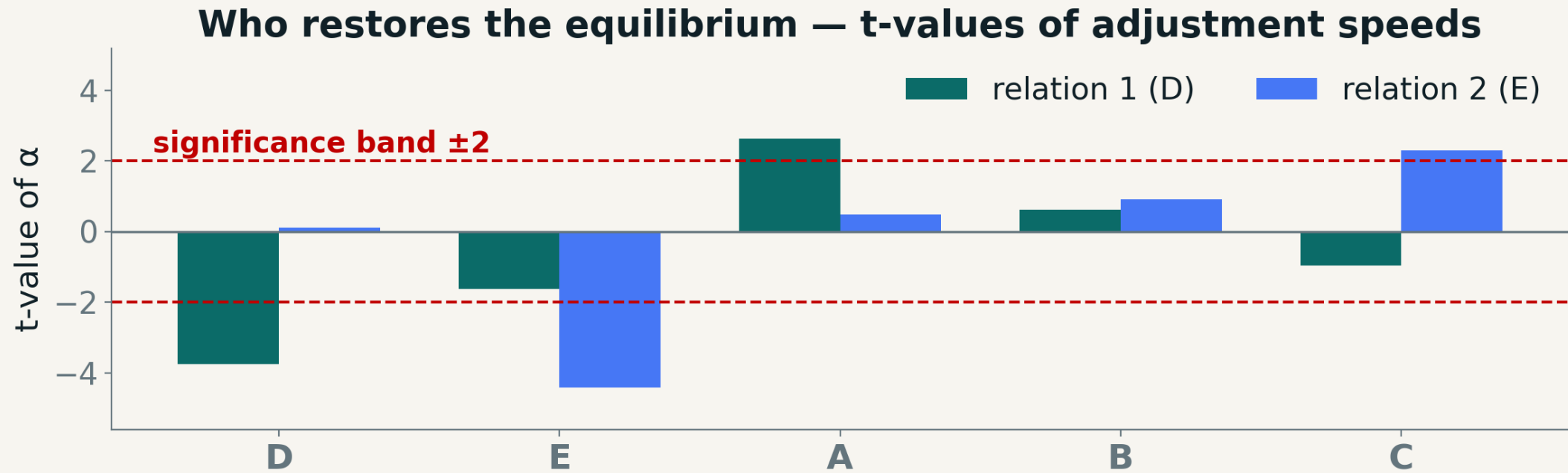
If X is weakly exogenous, a single equation suffices

$$H_0 : \lambda_X = 0 \implies X \text{ weakly exogenous}$$

Drop the contemporaneous difference; leaving it in contaminates the coefficient.

In cash and futures, the side that does not adjust is the one leading price discovery.

Five-asset VECM — the α matrix splits the roles



D restores relation 1 and E restores relation 2; B is insignificant in both columns.

※ VECM($k_{ar_diff} = 1$, rank = 2), T = 600. Bars are t-values of the adjustment coefficients

Three rules for reading the α matrix

variable	α on relation 1 (t)	α on relation 2 (t)	verdict
D	-0.111 (-3.76)	0.004 (0.12)	adjusts to relation 1
E	-0.057 (-1.63)	-0.160 (-4.42)	adjusts to relation 2
A	0.094 (2.63)	0.018 (0.49)	partly reacts to relation 1
B	0.024 (0.62)	0.037 (0.91)	weakly exogenous
C	-0.032 (-0.95)	0.081 (2.31)	partly reacts to relation 2

Positive α on A and C is not divergence — their β coefficients are negative.

ECM is used well beyond trading

FORECASTING

keeps the long-run tie

no drifting forecasts

IMPULSE RESPONSE

permanent vs transitory

common trends remain

price discovery

information share

the quiet side leads

MACRO USE

Testing theory

test theory by restricting β

LIMIT

threshold effects

threshold ECM with trading costs

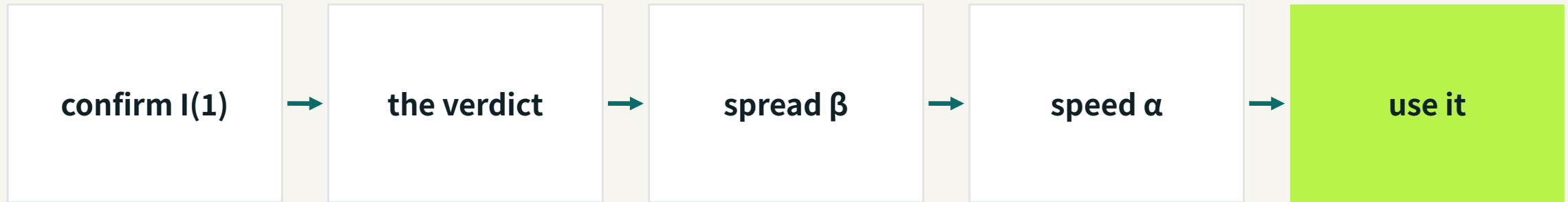
Hasbrouck's information share and the Gonzalo–Granger split both start from α and β .

Three tools, three faces of the same fact

crit	Engle–Granger	Johansen	ECM · VECM
question	is there cointegration	how many are there	how does it close
scope	two assets	three or more	everything after the test
output	one residual	r spreads	speed α
sensitive to	critical values, sides	lags, terms, ID	lags, structural breaks

Two assets: EG. Three or more: Johansen. After the verdict, always the ECM.

The whole flow, one more time



WHAT REMAINS

Spread and α

**WHERE READING
BEGINS**

The test is a rite of passage; what you actually use is the spread and the speed.

Prices wander apart.

The spread comes back.

α tells you the speed.

**From cointegration
to error correction**

※ Next: threshold ECM, permanent-transitory decomposition, restrictions on β